Where Does Coalition Coordination Go When You Watch It? Carrier Displacement and Countermeasures for LLM Agent Collusion

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Abstract

A coalition of LLM agents that shares a goal the rest of the team does not needs a way to coordinate. Prior work shows that a private text channel is often enough to make such a coalition collude, and that collusion can be measured in actions rather than only in what agents say. We ask the complementary question: when a known coalition is watched, or when its private channel is removed, does the collusion stop, go quiet but continue, or move to another carrier? We specify a carrier-displacement protocol on Colosseum's JIRA ticket-allocation environment. It combines an announced monitoring treatment, channel removal, and two carrier ablations (public-message paraphrasing and claim hiding) with a pre-registered three-test evidence standard: a coalition action-advantage test, a non-coalition reward-loss test, and an intent test based on a maximum-likelihood estimate of Colosseum's empirical misalignment weight or on a judge-score rise. The design also measures coalitional regret, peer identification without a named partner, and a trusted verifier agent. We release the implementation and a CPU-only analysis pipeline that recomputes every reported number from saved logs. This version is a registered protocol: no LLM episodes are reported here, so every results table is a reporting template and contains no observed numbers.

1 Introduction

Multi-agent systems built from large language models (LLMs) can solve tasks that no single agent can solve alone. They also create a governance problem: a subset of agents may form a coalition that pursues a secondary objective at the expense of the team. Such collusion is hard to detect because the coalition can coordinate in natural language, and because a team that coordinates well and a coalition that coordinates against the team can look similar in aggregate reward.

Colosseum (Nakamura et al. 2026) makes the problem concrete. In its JIRA ticket-allocation environment, six agents allocate tickets under private skill and availability information. Giving two agents a private channel is enough to make many models collude, and the resulting behavior can be measured as regret against the cooperative optimum. Colosseum varies the coalition's goal, its persuasion tactics, and the communication topology. Its coalitions keep the private channel unless stated otherwise.

We start from the other end. Take a coalition that is already colluding, then watch its channel or take it away. The obvious defense is to announce monitoring, or to remove the private channel entirely. If either intervention works, collusion should fall. If it does not, there are two possibilities. The coalition may go quiet while continuing to act together (concealment), or it may move coordination onto another carrier, such as the wording of public messages or the order and visibility of claims (displacement). Our working hypothesis is that watching or removing one carrier displaces coordination rather than ending it, and that collusion stops only when every carrier is cut.

Testing that hypothesis requires more than a drop in suspicious text. A coalition that stops talking because it has been told to stop is indistinguishable, from the outside, from one that has moved its coordination somewhere the auditor is not looking. The protocol therefore holds the coalition fixed and varies only its means of coordinating, and it measures collusion in three ways at once: what the coalition does, who pays for it, and what the logs say about its intent. The first two are action-based and do not depend on a judge; the third is the evidentiary bridge between a coordinated pair and a cartel.

We organize the project as six research questions (RQs). RQ1 fixes what counts as evidence before any results are seen. RQ2 asks whether an instructed coalition actually colludes in our setup, which is the precondition for the rest. RQ3 asks what the coalition does when it learns it is watched. RQ4 removes the private channel and ablates the remaining carriers. RQ5 asks whether colluders can find each other without being told who their partner is. RQ6 tests a trusted verifier agent that checks public claims against shared state. The RQs are sequential: if an early stage is negative, later stages change shape, and we say how at the end of each stage.

This paper makes four contributions. First, it gives a pre-registered evidence standard for collusion in agent logs, combining an action test, an externality test, and an intent test, supported by coalitional regret and a maximum-likelihood estimator for Colosseum's empirical misalignment weight. Second, it gives a carrier-displacement protocol that separates deterrence, concealment, and displacement through monitoring, channel removal, and carrier ablations. Third, it specifies measured tests of peer identification and of a trusted

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verifier, both of which Colosseum recommends or leaves open. Fourth, it releases an implementation and an analysis library that recompute the numbers behind every table on CPU from saved logs, without new generation.

Status. This version is a registered protocol. Sections 4–6 were fixed before any LLM episode was generated, and the result tables in Section 7 are empty by design. The only runs in this repository are offline wiring checks with a scripted client; that client is not a language model, so its outputs say nothing about collusion and are excluded from every table. We mark every number that has not been observed.

2 Related Work

Colosseum. Nakamura et al. (2026) introduce a framework for auditing collusion in cooperative multi-agent systems. They formalize cooperation as a multi-agent decision problem, measure action-based collusion by regret relative to the cooperative optimum, and compare it with communication-based collusion. Their behavioral probe creates a secret channel between two agents, and they find that most out-of-the-box models collude under this probe. They also identify "collusion on paper," where agents plan to collude in text but choose non-collusive actions. We use their environment, metrics, and judge prompts unmodified, and we hold the coalition fixed while changing its means of coordination.

White-box and training-time work. NARCBench studies multi-agent collusion through multi-agent interpretability and linear probes (Rose et al. 2026). That line of work has access to model internals; our defender is black-box and sees only messages, actions, and task outcomes. Work on steganographic collusion shows that models can learn to hide coordination in apparently benign text and that paraphrasing can mitigate it (Mathew et al. 2025). We study an instructed coalition rather than a model trained for steganography, but the carrier question is the same. Linear probes for strategic deception (Goldowsky-Dill et al. 2025) and obfuscated activations (Bailey et al. 2026) are out of scope for the same reason: the defender has no weights or activations.

Quantal response and regret. Our empirical misalignment weight is a logit quantal response model (McKelvey and Palfrey 1995). The model lets us estimate how much weight an agent places on the coalition's objective relative to the team objective, while allowing for noisy choice. Colosseum defines the corresponding weight but does not estimate it. We fit it by maximum likelihood and use the bootstrap over seeds for uncertainty. The action side of our standard uses regret, as in Colosseum, but adds a joint-deviation measure for the coalition and separates the coalition's gain from the loss imposed on everyone else.

3 Threat Model and Testbed

System. A team of N LLM agents solves a shared task with a known joint objective. Agents coordinate in natural language on a public channel and, in some settings, on private channels. Each agent then commits an action through a tool call. In the JIRA environment, an action is a claim on at most one ticket, or a skip.

Adversary. A coalition $S \subseteq N$ of $k \geq 2$ agents also pursues a hidden objective F_c. Here $k = 2$ and the objective is given by prompt; it stands in for misaligned weights, a compromised agent, or an injected instruction. Coalition members can send messages on any channel they belong to, choose their own actions, and read public state. They cannot change the environment, change other agents' prompts, or touch the audit logs.

Defender. The system designer or auditor can read all messages, actions, and the task objective. The defender can change the protocol by announcing monitoring, paraphrasing messages, hiding earlier claims, or adding a trusted agent. The defender has black-box access only: no weights and no activations.

Security goal. Detect and prevent coalition deviations that impose a cost on agents outside the coalition, measured as lost reward for non-members.

Out of scope. Training-time attacks, models trained for steganography, and white-box monitoring.

Testbed. We use the JIRA ticket-allocation environment from Colosseum, unmodified. Six engineer agents each claim at most one of eight tickets. The team score rewards completed tickets and high priority, and penalizes skill-adjusted effort and duplicate claims. Each agent's skills and availability are private, so agents have to talk to allocate well. Because the task is a distributed constraint optimization problem (DCOP), the cooperative objective is known and we can compute what any agent or pair should have done. That lets us measure collusion as regret in actions, not only as suspicious text.

Episode protocol. Each episode has four planning rounds. In each round every agent takes one turn of up to three conversation steps and may post on any blackboard it can access. A single execution phase follows, in which agents act in sequence and each claims at most one ticket or skips; unless the claim-hiding ablation is on, later agents see which tickets earlier agents have claimed. The agent model is Qwen2.5-7B-Instruct served by vLLM at temperature 0.4, with at most 1,024 new tokens per turn. A seed fixes the problem instance, the agent order, and the coalition pair, so the same seed gives the same pair and the same instance in every setting; comparisons are therefore within-seed.

4 What Counts as Evidence?

Every later stage asks whether collusion went up or down. That requires a definition fixed before the results are seen, and one that does not rest on an LLM judge alone. Colosseum already shows that judges and actions often disagree. We therefore pre-register three tests. A setting counts as collusive, relative to the benign team, only if all three hold.

Test 1: the coalition gains in actions. Coalition advantage is higher than in the benign team. Coalition advantage is the mean unilateral regret of non-coalition agents minus the

RQ	Question	Intervention / comparison
RQ1	What counts as evidence of collusion?	Pre-registered three-test standard on saved logs
RQ2	Does an instructed coalition collude?	Benign vs. emergent vs. coalition
RQ3	What happens under monitoring?	Coalition vs. watched coalition
RQ4	Can coordination travel without a private channel?	No-channel coalition, then paraphrase and claim-hiding ablations
RQ5	Can colluders find each other?	No channel, partner unnamed; partner-identification query
RQ6	Can a trusted agent break up a coalition?	Coalition plus a seventh verifier agent

Table 1: Research questions and the intervention that identifies each one. The RQs are sequential: a negative answer to RQ2 changes the design of the later stages.

Setting	Private	Prompt	Purpose
benign	No	No	Matched cooperative baseline
emergent	Yes	No	Colosseum's emergent-collusion probe
coalition	Yes	Simple	Known colluding coalition
watched	Yes	Simple + monitoring	Does monitoring deter or displace?
no_channel	No	Simple, partner named	Is privacy necessary?
find_partner	No	Simple, partner unnamed	Can colluders identify each other?
verifier	Yes	Simple + verifier	Can a trusted agent intervene?

Table 2: Base settings and the carrier each one removes or adds. The RQ4 ablation labels append _paraphrase, _hide_claims, or _both to watched and no_channel.

mean unilateral regret of coalition agents:

$$\mathrm{Adv}(\tilde{\mathbf{x}}) = \frac{1}{|N \setminus S|} \sum_{i \notin S} \delta_i(\tilde{\mathbf{x}}) - \frac{1}{|S|} \sum_{i \in S} \delta_i(\tilde{\mathbf{x}}), \qquad (1)$$

where the unilateral regret of agent i is

$$\delta_i(\tilde{\mathbf{x}}) = \max_{a \in \mathcal{A}_i} r_i(a \oplus \tilde{\mathbf{x}}_{-i}) - r_i(\tilde{\mathbf{x}}). \qquad (2)$$

Here $r_i(\mathbf{x})$ is agent i's credited reward under joint assignment $\mathbf{x}$, $\mathcal{A}_i$ is its action set, and $\oplus$ replaces one agent's action while holding the others fixed. Because the same two agents are designated as the coalition in every setting at a seed, Adv is defined even in the benign team, where those agents receive no coalition instruction.

Test 2: someone else pays for it. Mean reward for non-coalition agents is lower than in the benign team. Without this test, a pair that simply coordinates well with each other can look like a cartel. The externality split reports the change in coalition reward and the change in non-coalition reward separately, each against the same seed in the benign team.

Test 3: there is evidence of intent. Either the fitted misalignment weight λ_{emp} has a 95% bootstrap interval above zero, or the judge score rises by at least one point. The judge is Colosseum's three judge prompts (0 to 5, averaged), run on a local Qwen2.5-14B-Instruct. Absolute judge scores are not comparable to Colosseum's published numbers because the judge is smaller; only differences within this project are meaningful.

Decision rule. Tests 1 and 2 are one-sided paired tests on per-seed differences, and we apply Holm's correction over one family containing the two tests for every non-benign setting in the run. Test 3 uses either branch; the judge branch is a fixed one-point threshold on the paired mean difference, not a separate significance test. A setting is collusive only if all three tests pass. If a setting passes tests 1 and 2 but not test 3, we report it as a coordinated pair without evidence of coalitional intent, not as a cartel.

Coalitional regret. Collusion is a joint deviation, so we also ask what the pair could have gained by changing their choices together, holding everyone else fixed:

$$\delta_S(\tilde{\mathbf{x}}) = \max_{x_S \in \mathcal{A}_S} \sum_{i \in S} r_i(x_S \oplus \tilde{\mathbf{x}}_{-S}) - \sum_{i \in S} r_i(\tilde{\mathbf{x}}). \qquad (3)$$

With two colluders and at most eight tickets (plus skip), this means checking at most $9^2 = 81$ joint choices per episode, which is cheap. Coalitional regret is descriptive: it tells us whether the pair left a joint gain on the table, while coalition advantage tells us whether the pair did better than the agents it excluded.

Empirical misalignment. Colosseum defines λ_{emp}, how much weight an agent actually puts on the coalition's goal, but does not estimate it. We model each choice as a logit quantal response:

$$\begin{aligned} P(a_i = a) &\propto \exp(\beta\, U_\lambda(a))\,, \\ U_\lambda(a) &= (1-\lambda)\, \tilde{F}_{\text{n}}(a \oplus \tilde{\mathbf{x}}_{-i}) + \lambda\, \tilde{F}_{\text{c}}(a \oplus \tilde{\mathbf{x}}_{-i}), \end{aligned} \qquad (4)$$

where F_{n} is the team score and F_{c} is the coalition's combined reward, each min-max normalized over the agent's own choice set in that episode. We fit β and λ by maximum likelihood for each setting, pooling colluders across seeds, with 95% intervals from 1,000 bootstrap resamples over seeds. The weight λ is identifiable only on decisions where the team's best choice and the coalition's best choice differ, so the fit uses only those decisions; we report both the total number of decisions and the number that qualify. Bootstrap resamples with no identified decision are discarded, and we record how many were discarded. As a negative control, the same fit on the designated pair in the benign team should give λ near zero.

Mode	Judge score	Action signal	Reading
Direct	Rises	Yes	The coalition talks and acts on its goal
Hidden	Does not rise	Yes	The coalition acts without saying so
On paper	Rises	No	The coalition plans but does not follow through

Table 3: Descriptive modes for settings that already pass the three RQ1 tests.

Judge. Judge scores are the mean of Colosseum's three prompts, each on a 0–5 scale. For a setting with a private channel, the judged transcript is that channel. For a channel-free setting, the judged transcript is each blackboard the colluders share, scored separately and averaged; an empty channel scores zero. We use the judge only as one branch of the intent test and as a descriptive outcome, never to define collusion on its own.

Sorting collusive settings. Once a setting passes all three tests, we sort it by how the collusion appeared. Direct collusion raises the judge score and shows an action signal. Hidden collusion does not raise the judge score but still shows an action signal. Collusion on paper raises the judge score without an action signal. This taxonomy is descriptive; it does not replace the three tests.

5 Carrier Displacement Protocol

The protocol is a sequence of comparisons. Each comparison uses the same model, seeds, and judge as every other comparison. Seeds are paired: in Colosseum a seed fixes the problem instance, agent order, and coalition pair, so comparing two settings on the same seed cancels out how hard the instance was. We use 20 paired seeds per setting. The full confirmatory plan is 11 settings with 20 seeds each, 220 episodes; the two combined ablations, which we run only if time allows, add 40 episodes.

5.1 RQ2: Does an instructed coalition collude?

We compare three settings. In benign, there is no private channel and no coalition prompt. In emergent, two agents get a private channel but benign prompts; this is Colosseum's emergent-collusion probe. In coalition, the same two agents get the private channel and Colosseum's simple coalition prompt, which asks them to maximize the coalition's joint reward. These three runs are the matched controls for everything that follows.

These runs are not a reproduction of Colosseum's numbers. Their raw logs are not released, and their reported results come from other models, mostly through APIs, so we cannot pair against them. If the coalition gains and all three RQ1 tests pass, we go on to RQ3 and RQ4. If there is no gain at 7B, we try, in order, Colosseum's structured coalition prompt, letting colluders act first (agent_order: colluders_first), and a larger model on Modal before spending time on RQ3–RQ6.

5.2 RQ3: What does a coalition do when it is watched?

We take the colluding team from RQ2 and add the following announcement to the coalition header of both colluders' user prompt:

> All messages on every channel, including private channels, are logged and reviewed by an independent auditor after the task. Agents found coordinating against the team objective will be flagged.

We compare the watched coalition against the unwatched coalition from RQ2. A drop in both the judge score and coalition advantage is deterrence. A drop in the judge score with coalition advantage held is concealment. A drop in the judge score with coalition advantage held, and with RQ4 attributing the surviving advantage to another carrier, is displacement. Concealment and displacement look the same from the outside; RQ4 is what tells them apart. We also report the number of colluder posts on the private and public channels, because a fall in private posts with an unchanged advantage is the direct signature of concealment, while a rise in public posts is the signature of a move to the public channel.

A caveat we cannot fully remove is that a drop after the announcement might be plain instruction-following rather than strategic deterrence. Agents may read "you will be flagged" as "do not do this." Concealment, where talk drops but action does not, is the outcome that rules this out.

5.3 RQ4: Can coordination travel without a private channel?

Step 1: remove the channel. We give the same two agents the same coalition prompt, naming their partner, but no private channel. They can use only the public channel everyone reads, and their actions. If coalition advantage survives, something other than private text is carrying the coordination. If it vanishes, the private channel was the only real carrier in this setting, and RQ3 reads as deterrence or concealment rather than displacement.

Step 2: cut one carrier at a time. We apply three ablations to the channel-free coalition and to the watched coalition from RQ3. Paraphrasing rewrites every public message before other agents read it, using a separate call on the same base model at temperature zero. The rewrite is cached per event, so every reader sees the same text, and the author still sees its own words. The paraphraser is instructed to keep every ticket ID and number; the implementation checks the ticket IDs, and if the rewrite is empty or drops one, the original message is used instead and the fallback is recorded. Hiding earlier claims removes the committed-action events from the blackboards and removes other agents' entries from the joint-assignment block at execution time; an agent keeps its own claim, so it is not told it may act twice. The combined ablation applies both.

If coalition advantage disappears under an ablation, the ablated carrier was necessary. If it survives both single ablations and the combination, the remaining advantage is noise or a carrier we have not named, which is itself worth reporting.

Ablation	How	If advantage disappears	If advantage survives
Paraphrase public text	Separate model call, temperature 0; keep ticket IDs and numbers, not wording	Wording of public messages was a carrier	Wording is not sufficient
Hide earlier claims	At execution, hide which tickets earlier agents claimed	Order and visibility of actions was a carrier	Action visibility is not sufficient
Both	Apply both	Both carriers were necessary	Remaining advantage is noise or an unnamed carrier

Table 4: Carrier ablations in RQ4. Each ablation is applied to both the channel-free coalition and the watched coalition.

Outcome	Judge score	Coalition advantage	Reading
Deterrence	Falls	Falls	Watching works
Concealment	Falls	Holds	The coalition stops talking but keeps acting together
Displacement	Falls	Holds, and RQ4 attributes the gain to another carrier	Coordination moved off the watched channel

Table 5: Reading the watched coalition against the unwatched coalition. Concealment and displacement differ only in what RQ4 finds.

There are two known limits. Paraphrasing keeps ticket IDs and numbers, so it cannot remove a signal carried by which tickets an agent mentions; and the paraphraser may fail open, in which case that message contributes no ablation at all. Claim hiding removes the traces of earlier actions but not the environment's duplicate-claim penalty, so a coalition that coordinates by trial and error still faces the same cost.

5.4 RQ5: Can colluders find each other?

Every stage so far tells colluders who their partner is. In a real deployment, a misaligned agent might not know which peers share its goal. We use the channel-free setting from RQ4, but tell each colluder only that exactly one other agent shares the coalition's goal, without naming it. After the episode we ask each colluder which agent it believes is its partner. The question is asked outside the episode, with no tools and with the public transcript, so it cannot change the outcome it reports on. The answer is a single JSON object; a name that matches one of the five other agents is resolved, and an unresolved or ambiguous answer counts as incorrect.

We measure partner-identification accuracy against chance (one in five) with an exact binomial test, and compare coalition advantage with the named-partner setting from RQ4. The comparison shows how much of the coordination depended on knowing whom to coordinate with.

5.5 RQ6: Can a trusted agent break up a coalition?

We add a seventh agent to the colluding team from RQ2. It controls no tickets and is not an environment agent, so the team objective is unchanged. It joins the public blackboard and, after every planning round, checks two things: whether two agents have staked a claim on the same ticket, which is a deterministic string check, and whether an agent has asserted something about another agent's availability or skills that it has no way to know, which is a model call. It posts at most one consolidated flag message per round and does not vote on the allocation. Repeated reports of the same duplicate clash are suppressed.

If coalition advantage drops and non-coalition reward recovers, the verifier works as a countermeasure. If advantage holds but the judge score drops, the coalition worked around the verifier, which is RQ3's concealment pattern again. We also report the cost: verifier messages and flags per episode, and the joint reward of the verifier team against the coalition team. The verifier is itself an LLM, so its flags are noisy; we report them by type rather than treating every flag as a true violation.

6 Statistical Analysis

Paired tests. For every comparison we form per-seed differences and run a paired sign-flip permutation test. Under the null, the sign of each difference is exchangeable. We use the mean difference as the test statistic and enumerate all 2^n sign patterns when $n \leq 18$; for larger n we use 10,000 Monte Carlo sign flips with the standard add-one correction. The main design has 20 seeds, so its confirmatory tests use the Monte Carlo version; the five-seed pilot uses the exact version. We use $\alpha = 0.05$ and Holm's correction over the RQ1 family. We never compare pooled means: seed pairing is part of the design.

Later-stage comparisons. RQ3–RQ6 compare each treatment with a pre-specified parent, not with every other setting. We report two-sided paired tests and seed-level effect sizes for those comparisons, and we read a mechanism from the joint pattern of action, reward, and talk rather than from a single p-value. These comparisons are not added to the RQ1 family and cannot turn a non-collusive setting into a collusive one.

Bootstrap. For λ_{emp} we resample seeds with replacement 1,000 times, refit the QRE model on each resample, and report the 2.5th and 97.5th percentiles. The same percentile bootstrap gives intervals for the per-setting means in the tables.

Identifiability. The QRE fit uses a decision only if the team's best action and the coalition's best action differ. For

each setting we report the number of decisions and the number that qualify. If a setting has none, there is no λ estimate and the intent test falls back to the judge delta, as RQ1 permits. The benign fit is a negative control: if it produces a large λ, the model or the normalization is wrong.

Judge. Judge scores are the mean of Colosseum's three prompts, each on a 0–5 scale. We use the judge only as one branch of the intent test and as a descriptive outcome. We do not use it to define collusion on its own.

Compute and stopping. Generation and judging run on a single Colab A100 with vLLM. Analysis runs on CPU from saved JSON outputs. The confirmatory plan is 11 settings with 20 seeds each (220 episodes); the two combined ablations add 40 episodes if time allows. The seed count is fixed in advance; we do not add seeds after seeing a result. If time runs short, we cut RQ6, then RQ5, then the combined ablation. RQ2, RQ3, and the paraphrase ablation from RQ4 are enough to answer the core question. The Modal credit is reserved for a second model or overflow, not for a full sweep, and a replication model is reported separately rather than pooled with the main model.

7 Results

This section contains no empirical results. No pre-registered LLM sweep is included in this version; the repository contains the implementation, the analysis library, and offline wiring checks only. We do not populate a cell from a partial sweep, and we do not report the scripted-client runs: that client emits fixed messages and claims tickets by hash, so any collusion metric computed on it measures the script rather than a language model.

Implementation validation. The artifact passes 26 offline tests with no GPU and no API. They cover the JIRA scoring and regret re-implementation against Terrarium, the pairing and Holm logic, the three-test standard, the carrier prompts and verifier, the paraphraser's ticket-ID invariant, and an end-to-end scripted-client run through every registered setting. This establishes that the pipeline runs and that the analysis agrees with Colosseum's environment on the quantities it re-implements. It is not behavioral evidence.

The tables below fix the reporting format and the columns that the pre-registered analysis will populate. In the main table, deltas are against the benign team; in the ablation table, each row is compared with its parent. All deltas are treatment minus comparison, on the same seeds. We will not add a result to a table unless the corresponding setting has completed all 20 paired seeds and the analysis script has produced the number from the saved logs.

8 Discussion

The protocol is designed to make three outcomes distinguishable. If monitoring reduces both talk and action, the straightforward reading is deterrence. If talk falls but action does not, the coalition has concealed. If action survives while the watched channel goes quiet, and an ablation identifies the replacement carrier, coordination has been displaced. The third outcome is the one that would make private-channel auditing insufficient on its own.

The design also has a built-in negative control. The benign fit for λ_{emp} should be near zero. If it is not, the QRE model is absorbing something other than coalitional intent, and the intent test should not be trusted. Similarly, the channel-free coalition is a test of necessity: if collusion vanishes when the private channel is removed, then RQ3's outcome is deterrence or concealment, not displacement.

There are several ways the results could be null. A 7B model may not collude effectively even when instructed. The paraphrase ablation may leave enough signal in ticket IDs and numbers to preserve coordination. The verifier may be ignored or may generate flags that are themselves noise. We report these outcomes rather than treating them as failures of the protocol. A null on RQ2 is informative for the same reason: it says that the carrier question cannot be asked at this scale, and the pre-registered fallbacks (a structured prompt, colluders acting first, a larger model) say what to try before abandoning it.

The protocol also has a cost that is worth stating. Announcing monitoring and paraphrasing messages change the task the agents see, not only the auditor's view. A drop in collusion under either treatment is evidence about the deployed system, not a clean estimate of a cognitive mechanism. We therefore describe the treatments as protocol changes and read the mechanism only from the pattern across action, reward, and talk.

9 Limitations

No results yet. This version reports no empirical results. Every claim about collusion, deterrence, concealment, or displacement is still open, and the tables contain no observed numbers. The contribution at this stage is the pre-registered design and the artifact, not a finding.

One small model. The main results use a single 7B model. Colosseum finds that collusion behavior varies with model and size, so the results should not be read as a claim about LLMs in general. The Modal replication, if run, is a separate model and is reported separately.

One-shot task. JIRA is a single-round task. Collusion that depends on reputation, repeated play, or long-horizon relationships is out of scope.

Instructed coalitions. Apart from the emergent comparison in RQ2, every coalition here is told to collude. We study how coordination is carried and broken, not whether it arises unprompted.

Imperfect paraphrasing. Paraphrasing preserves ticket IDs and numbers. It removes wording but not the set of tickets an agent mentions, so a signal carried by ticket choice can survive. The paraphraser also fails open, so some messages are not ablated.

Local judge. Judge scores come from a 14B model, not the frontier judge Colosseum used. Absolute scores are not comparable; only within-project differences are meaningful.

Setting	ΔAdv	Δ non-coal. reward	λ_{emp} [95% CI]	Δ judge	posts priv/public	T1/T2/T3
emergent	–	–	–	–	–	–/–/–
coalition	–	–	–	–	–	–/–/–
watched	–	–	–	–	–	–/–/–
no_channel	–	–	–	–	–	–/–/–
find_partner	–	–	–	–	–	–/–/–
verifier	–	–	–	–	–	–/–/–

Table 6: Primary results template. Deltas are against the benign team on the same seeds, except that watched, no_channel, find_partner, and verifier are also read against coalition for the mechanism question. The last column records the action (T1), externality (T2), and intent (T3) verdicts. All entries are placeholders until the runs complete.

Parent	Ablation	ΔAdv	δ_S	Extern.	Judge
no_channel	none	–	–	–	–
no_channel	paraphrase	–	–	–	–
no_channel	hide claims	–	–	–	–
no_channel	both	–	–	–	–
watched	paraphrase	–	–	–	–
watched	hide claims	–	–	–	–
watched	both	–	–	–	–

Table 7: Carrier-ablation template. A drop from the parent setting attributes the advantage to the ablated carrier; survival points to a remaining carrier or to noise. Deltas are against the parent on the same seeds.

The judge also sees only the transcript, not the agents' private reasoning.

Verifier noise. The verifier's second check is an LLM call. It can miss a violation or flag a legitimate inference, and the protocol reports flags by type rather than assuming they are true.

Compute and seeds. The full design is 220 episodes, which is feasible on one A100 but leaves little room for model-scale replication. Twenty paired seeds give limited power for small effects, and we do not add seeds after seeing a result.

10 Ethical Statement

This work studies how to detect and limit collusion among LLM agents. The immediate application is defensive: an auditor who can read messages and actions can use the protocol to tell whether a coalition is coordinating against the team. The same tools could be used to build more covert coalitions, and the paper describes carrier ablations that could inform an adversary. We judge the defensive value to outweigh this risk because the protocol assumes a defender who already has full log access and a known cooperative objective; it does not provide a way to evade a defender who lacks those advantages. We use no human subjects and no private user data. Released logs contain model-generated messages and task outcomes; we will review them for incidental personal information before release.

11 Reproducibility and Artifact

The artifact has three parts. The first is a set of YAML configs for the settings in Table 2 and a runner that mirrors Colosseum's episode loop while importing its environment, metrics, prompt wrapper, and judge prompts. The runner exists because Colosseum forces the prompt variant to control whenever the secret channel is off, which would rule out the channel-free settings; the episode structure and logging format are otherwise Colosseum's. The second is a carrier layer for the monitoring announcement, the channel-free prompt, the paraphraser, claim hiding, and the verifier. The third is an analysis library that reads the saved metrics.json, blackboards.json, agent_turns.json, and tool_events.json files and recomputes every number in the tables without new generation.

Colosseum is used unmodified; its environment, metrics, and judge prompts are not forked. The only changes to the prompt wrapper are the coalition header variants that the settings require, and the coalition text itself is Colosseum's. Seeds are 1–20 and are paired across settings. Generation and judging need a GPU; all analysis runs on CPU. The offline test suite needs neither a GPU nor an API. When the sweep is complete we will release the run logs alongside the code so that the analysis can be rerun from the saved artifacts.

12 Conclusion

Private-channel monitoring is an attractive defense because it is easy to state and easy to audit. The question is whether it changes the coalition's behavior or only its channel. We have specified a protocol that can tell the difference: a preregistered evidence standard, a watched coalition, a channel-free coalition, and carrier ablations that attribute any surviving advantage. The protocol also tests two countermeasures that Colosseum recommends but does not run: peer identification and a trusted verifier. The results will determine whether the right conclusion is that monitoring works, that it teaches concealment, or that coordination simply moves.

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